

Lesson 2

Electric fields

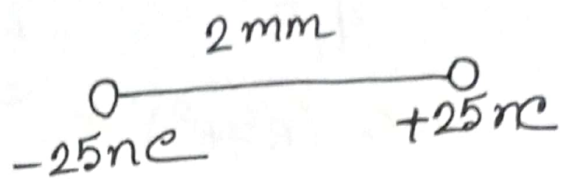
1. Given,

$$q = 25 \text{ nC}$$

$$= 25 \times 10^{-9} \text{ C}$$

$$d = 2 \text{ mm}$$

$$= 0.002 \text{ m}$$



$\therefore$ Electric dipole moment $p = qd$

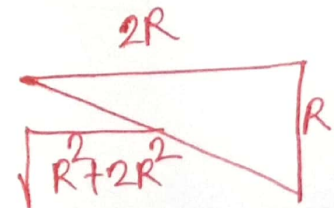
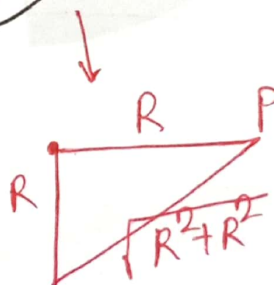
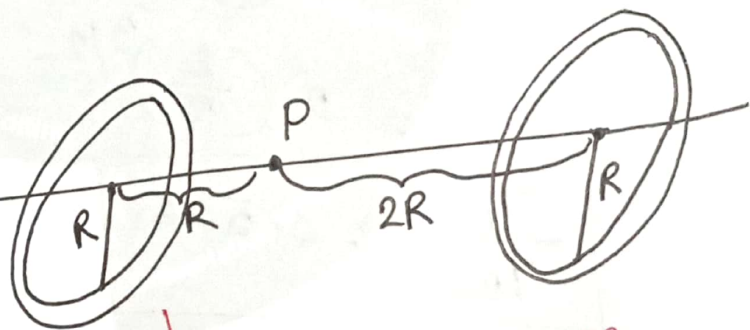
$$= (25 \times 10^{-9}) \times (0.002)$$

$$= 5 \times 10^{-11} \text{ C.m}$$

2. We know,

$$E = \frac{qz}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}}$$

$$\therefore E_{\text{left}} = \frac{q_1 R}{4\pi\epsilon_0 (R^2 + R^2)^{3/2}}$$



$$E_{\text{right}} = \frac{q_2 (2R)}{4\pi\epsilon_0 \{(2R)^2 + R^2\}^{3/2}}$$



Centre for Medical Research & Development (CMRD)

$$E_{\text{Left}} = E_{\text{Right}}$$

$$\frac{q_1 R}{4\pi\epsilon_0 (R^2 + R^2)^{3/2}} = \frac{q_2 (2R)}{4\pi\epsilon_0 [(2R)^2 + R^2]^{3/2}}$$

$$\Rightarrow \frac{q_1}{(2R^2)^{3/2}} = \frac{2q_2}{(5R^2)^{3/2}}$$

$$\Rightarrow \frac{q_1}{q_2} = \frac{2(2R^2)^{3/2}}{(5R^2)^{3/2}}$$

$$= 2 \left(\frac{2R^2}{5R^2} \right)^{3/2}$$

$$= 2 \left(\frac{2}{5} \right)^{3/2}$$

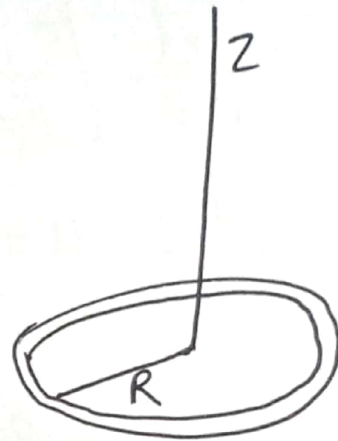
$$\boxed{\frac{q_1}{q_2} \approx 0.506}$$

13

$$E = \frac{qz}{4\pi\epsilon_0 (R^2 + z^2)^{3/2}}$$

(a) $z=0$
 $\Rightarrow E=0$

The field vanishes
at the center.



(b) $z = \infty$; $\frac{1}{z^2} \rightarrow 0$
 $E=0$

(c) E maximum for what value of z?

$$\frac{dE}{dz} = 0$$

$$E = \frac{qz}{4\pi\epsilon_0 (R^2 + z^2)^{3/2}}$$

$$\frac{dE}{dz} = \frac{d}{dz} \left[\frac{qz}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} \right] = 0$$

$$\Rightarrow \frac{q}{4\pi\epsilon_0} \frac{d}{dz} \left[\frac{z}{(z^2 + R^2)^{3/2}} \right] = 0$$

$$\Rightarrow \frac{q}{4\pi\epsilon_0} \frac{R^2 - 2z^2}{(z^2 + R^2)^{5/2}} = 0$$

details in P.T.O

Centre for Medical Research & Development (CMRD)



$$\frac{d}{dz} \left[\frac{z}{(z^2+R^2)^{3/2}} \right] = 0$$

Let's $z = u$; $(z^2+R^2)^{3/2} = v$

We know, $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{d}{dx} u - u \frac{d}{dx} v}{v^2}$

$$\therefore \frac{d}{dz} \left[\frac{z}{(z^2+R^2)^{3/2}} \right] = 0$$

$$\Rightarrow \frac{(z^2+R^2)^{3/2} \frac{d}{dz} z - z \cdot \frac{d}{dz} (z^2+R^2)^{3/2}}{\{(z^2+R^2)^{3/2}\}^2} = 0$$

$$\Rightarrow \frac{(z^2+R^2)^{3/2} \cdot 1 - z \cdot \frac{3}{2} (z^2+R^2)^{1/2} \cdot 2z}{\{(z^2+R^2)^{3/2}\}^2} = 0$$

$$\Rightarrow \frac{(z^2+R^2) \sqrt{z^2+R^2} - 3z^2 \sqrt{z^2+R^2}}{(z^2+R^2)^3} = 0$$

$$\Rightarrow \frac{\sqrt{z^2+R^2} (z^2+R^2 - 3z^2)}{(z^2+R^2)^3} = 0$$

$$\Rightarrow \frac{-2z^2+R^2}{(z^2+R^2)^3 (z^2+R^2)^{-1/2}} = 0$$



$$\Rightarrow R^2 - 2z^2 = 0$$

$$\Rightarrow R^2 = 2z^2$$

$$\Rightarrow z^2 = \frac{R^2}{2}$$

$$\therefore z = \frac{R}{\sqrt{2}}$$

$$z = 0.707 R$$